

Emotion Design for Video Games: A Framework for Affective Interactivity

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Emotionally evocative games hold great potential for both industry and our basic understanding about human affect. The question of how best games elicit and respond to player emotion brings together research from psychology, design, and human-computer interaction (HCI). However, there is much confusion regarding the nature of emotions, emotional game-player interaction, and the process of creating emotional games. This article reviews important work from relevant fields to provide an interdisciplinary model that integrates current theories and highlights conceptual gaps for relevant fields. Based on this model, we propose a methodological framework that provides a step-by-step process of developing an emotion-adaptive game. This framework is open to different models of emotion, and its modular nature allows games to adapt to developments in the science of emotion and technological innovations in emotion detection. Together, the theoretical overview and framework can help reduce the confusion that surrounds creating and researching emotions in games.

CCS Concepts: • **Human-centered computing** → **Human computer interaction (HCI)**; **HCI theory, concepts and models**; *User models*; *Interaction design theory, concepts and paradigms*;

Additional Key Words and Phrases: Game design, emotions, affective games, affective computing, emotional design

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1 INTRODUCTION

People love to play video games for multiple reasons. Central among these lie emotional benefits [78] and the fulfillment of basic psychological needs [113]. The view of games as emotionally impactful art is gaining popularity as people begin to understand the potential and unique ways in which video games can evoke emotions [127]. In addition to their commercial appeal, emotionally evocative games also provide researchers in **human-computer interaction (HCI)** and psychology with new ways to learn about the relationships between humans and machines and, more broadly, the nature of emotion itself. Nevertheless, the design and development of such emotional games is a difficult process that is explored with a wide range of techniques and is based upon often

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conflicting theoretical perspectives. Building on our knowledge of game enjoyment [87], **player experience (PX)** design tries to use game elements to elicit engagement and rich emotional experiences [117]. The impact of PX design is dependent on multiple inter- and intra-personal factors relating to the player's emotions [100], which makes it necessary to view emotional impact not only as an attribute of game material but also as a consequence of the emotional interaction between the game and the player.

The design of affective games aims to address this problem by conceptualizing games as dynamic systems that adapt to players' emotions [62]. Following the definition by Gilleade et al. [48], these games measure emotions and adapt themselves primarily to assist in frustrating situations, keep the player engaged with appropriate challenges [e.g., 47, 83], or "emote" the player to elicit specific emotional reactions [e.g., 95, 134].

A growing body of research provides evidence for the benefits of such an approach. These benefits can be categorized in four main ways: (1) both mental and physical health benefits as a result of new adaptive training opportunities [2, 4, 19, 57, 69]; (2) more accessible games with similar experiences for every player [47]; (3) new gameplay opportunities, like body-response control games or affective user interfaces [122, 145]; and finally (4) more emotionally impactful, immersive, and therefore better games [63, 147].

There are, however, conceptual uncertainties about what affective games are and how they can be developed. For example, affective games were originally seen only as biofeedback games [9] and are often still associated with that history, although later definitions focus more generally on adaptivity to unconscious or conscious emotional data [48, 62]. Such definitions could arguably include games that are adaptive in certain areas, such as open-world games like *Red Dead Redemption II* [45] or *Cyberpunk 77* [105] that change their narrative and game world depending on in-game behavior or, indeed, all games with multiple difficulty settings. Nevertheless, affective games are mostly presented in the literature as a certain type of game that measures and reacts to primary emotional data [12]. This conceptualization limits game design decisions that might tap the aforementioned benefits of affective games.

In addition to the conceptualizations of the games themselves, there are also uncertainties regarding player emotion. Within modern psychological research, there are a variety of competing theories and open questions regarding emotion elicitation and expression. As a consequence, affective and emotional games are often based on emotion models that fit a specific problem, leading to inconsistent applications that differ in their fundamental assumptions about the nature of emotions. In a recent systematic review of high-quality evaluations of affect-adaptive video games, Croissant et al. [25] found that affective game studies applying psychological theories varied widely in how emotions were defined, conceptualized, labeled, and measured, often assuming mappings between game material, user input, and emotions based on very specific and outdated theories.

In summary, there is a need for high-level theoretical clarification of relevant concepts to achieve more consistent definitions about the use of emotions in video games. A clear understanding of relevant concepts will enable developers and researchers to analyze the emotional draw of successful existing games, research emotional phenomena using games, and potentially create targeted emotional video games. Therefore, this study will provide a brief review of relevant literature on current developments in emotion research, design research, and affective computing and will use this overview to develop a theoretical model for ongoing emotional interaction in video games. To bridge our theoretical knowledge with practical methods, we will furthermore present a methodological framework that will provide necessary steps for incorporating emotion within the design process. This study therefore provides both a theoretical overview and a practical guide to explore and cultivate our emotional connection to games.

2 EMOTION RESEARCH

2.1 Definition of Emotion

The first step in designing for emotions is understanding them on a conceptual level, which has been an ongoing scientific effort for centuries. The terms “affect” and “emotion” are often used interchangeably, especially in the area of applied games. Modern psychological research, however, views affect as an umbrella term, describing multiple possible affective states that can be differentiated through certain design features [41, 149], such as duration or intensity. Consequently, emotions are seen as episodic affective states [37, 43, 118]. This means that they have a fixed duration with varying emotion-specific onset and offset periods of a certain intensity [120]. Differently from other forms of affective states such as mood, stances, or attitudes, emotions are characterized by a rather short duration and a high (and rapidly changing) intensity [41]. One additional aspect is the strength of the connection to a triggering event. While affective states such as attitudes and particularly affective traits are very stable and independent from triggering events, emotion elicitation is based on some form of interaction with an event [41, 149]. Since emotions are short and rapidly changing and based on events, they can be considered especially impactful in moment-to-moment gameplay.

There are many theoretical models discussed in modern psychology that offer varied, detailed, and complex analyses of how emotions can be modeled or differentiated. As of yet, there is no consensus on the exact nature of emotions. Nevertheless, many applied emotion research studies use specific theoretical assumptions that are still debated. This has led to a wide variety of definitions in the literature, especially within HCI research, which makes it difficult to frame emotions in a consistent way. In order to create a model that is flexible in terms of contentious issues but still integrates current psychological knowledge, it is important to give an overview about points of both disagreement and agreement.

2.2 Common Contentious Aspects of Emotions

2.2.1 Dimensional vs. Discrete Emotions. Historically, the most important criterion of distinguishing emotions was the pleasantness-unpleasantness dimension [144]. Common scales, such as the **Positive and Negative Affect Scales (PANAS)** [140], reflect this dimension and are still being used to measure emotional states in an experimental or diagnostic setting. Building upon such a unidimensional model, Russell [111, 112] popularized the circumplex model of affect (Figure 1), which added a dimension for activation or arousal, giving the classification more depth, while also providing a means of distinguishing between graphical illustrations of differences between emotions with similar valence. Recent support for dimensional models of emotion relies upon evidence that patterns of neural activity [142] or peripheral physiological responses [15] map onto these two dimensions.

In contrast to dimensional models, discrete emotion models argue for specific and distinct emotion expressions and action motivations. They originate from the pioneering work of Charles Darwin and Phillip Prodger [27], and the modern description of discrete emotions has been popularized by the works of Tomkins [135] and Izard [68], which resulted in the common description of emotion categories, such as joy, sadness, anger, fear, or disgust. Neuropsychological circuit models mirror this idea by arguing for emotion-specific neural networks for certain emotions, which resulted from an evolutionary necessity [79, 98].

In HCI studies, especially in affective games, emotions are most commonly considered in a dimensional perspective and measured in their expression of valence or arousal [36]. For example, a recent systematic review of affective games by Robinson et al. [108] defined the term “affect” as the mapping between physiological states in Russell’s circumplex model [111] (e.g., arousal) and

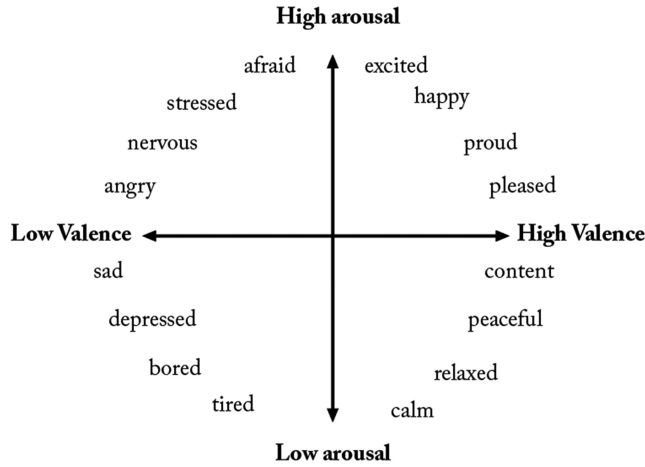


Fig. 1. Graphical illustration of the circumplex model, as described by [104].

emotional labels (e.g., relaxation), therefore considering physiological affective games by definition as dimensional. While this can be seen as a comparatively easy way to make affective data interpretable, many modern researchers acknowledge the complex adaptive nature of emotions. In a review of emotion measures by Mauss and Robinson [85], the author concludes that there are no clear mappings between measures (e.g., physiology) and emotions. Similarly strict inferences from dimensional models to distinct emotional states are considered a controversial process [56]. In fact, oversimplifying conceptual assumptions of either theoretical perspective could pose a risk to understanding a wide range of emotional phenomena [55], which makes it necessary to guide affective game studies and designers to not make unnecessary and controversial mapping assumptions.

2.2.2 Basic vs. Constructed Emotions. Similarly, a point of contention is the view of emotions as universal categories versus emotions as social constructions. Probably the most famous proponent of basic emotions is Ekman, who provided evidence of universal and therefore culture-independent facial expressions [38]. This has led to the view of emotions as fixed, biology-dictated constructs. Contemporary constructivist models of emotion, on the other hand, argue that although the “raw materials” of affect are innate, the emergence of a specific emotion is dependent on individual, situational, and culturally specific categorizations [8].

Again, theoretical assumptions are applied in different design and HCI studies, depending on the context. For example, games using cameras to capture facial expressions to control gameplay often base their classification on basic emotion models [e.g., 88, 122], while interactional affective computing models, like the one from Boehner et al. [11], propose methods to interpret emotions based on the situational context within a human-computer interaction and not based on their physiological expression.

2.2.3 Summary. There are more theoretical points of contention currently in discussion, including the question if certain emotion categories have specific physiological “fingerprints” or if these categories are rather context-specific, conceptual populations [126]. These different theoretical perspectives have an impact on the interpretation of emotion elicitation, general emotion function, and the proposed mechanisms that differentiate emotions [120].

Because fundamental aspects of emotions are still the subject of debate, it is difficult to make assumptions that are both immediately useful and resilient to changes in our understanding of

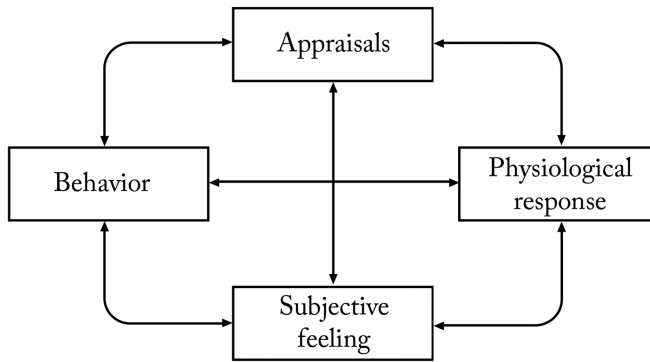


Fig. 2. Illustration of interactions between multiple emotion components.

emotions. While a given approach might seem expedient for a given design (e.g., dimensional models for biofeedback games), there is a risk of blindly assuming theoretical implications or disregarding important and subtle nuances of a given theory when modeling emotional reactions. Awareness and acknowledgment of a model's underlying assumptions will allow designers to anticipate higher-risk decisions and accommodate to future developments in the field. Moreover, focusing on theoretical assumptions that are shared across emotion models when developing emotional games provides the most flexibility, making the process resilient for future research developments and applicable for any theoretical perspective. With this in mind, the framework presented here relies heavily on key overlap between emotion models.

2.3 Theory Overlap

2.3.1 Emotions Consist of Multiple Components. Most theorists acknowledge that emotions have multiple components and that these components influence each other in various ways, as depicted in Figure 2. Commonly, emotion components include physiological, behavioral, and subjective (feeling) aspects. While there is debate about the number and nature of emotion components, it is generally assumed that emotions reflect patterns in componential expressions. Many modern theories regard these patterns as key emotion characteristics [49, 119], and some even argue that emotion measurement techniques should capture multiple components for a single emotion to reflect all relevant and distinct aspects [73].

2.3.2 Appraisal Plays an Important Role. Emotions result from triggering events, which might be internal (like thoughts or memories) or external (like situations). The notion of emotion appraisal is based on the assumption that emotions are caused by subjective evaluations of any event with regard to their personal significance and the organism's ability to cope [5, 76]. Generally, appraisal theories assume that an event is appraised on multiple variables, including goal relevance, goal congruence, certainty, coping potential, and agency [92]. It is important to note that it is not assumed that appraisal is a uniquely conscious or rule-based process. In fact, emotion elicitation is considered to be highly automatic [91], even though reflective and controlled processes play an important role [125]. This means appraisal includes different processing levels, e.g., automatic sensory-motor processing, automatic perceptual and associative processing, and non-automatic and rule-based conceptual processing [80]. It is generally believed that multiple pathways of top-down and bottom-up processes interact in the development of emotions [115], which means that appraisal leads to emotional expressions but is at the same time influenced by other components.

2.3.3 Emotions Are Context Specific. Building on the importance of appraisal, there is a high emphasis on the role of context in the development of emotions [42]. Following evidence for constructed emotion aspects, most modern researchers emphasize that the context of the emotional material plays an important role in the appraisal and expression of emotions [89]. For example, people sharing a close relationship show similar emotion reactions compared to strangers [14], and culture has been shown to have a big influence on the evaluation of emotional materials and emotions themselves [66]. Furthermore, applied emotion research have found that the environmental context of a situation is an important factor to elicit emotional reactions [75, 143]. For example, how empathy can be elicited with HCI design techniques depends on the used material (e.g., narrative, dialogue, or role-play) and the conceptualization of a user (e.g., as a participant or observer) [143]. In other words, appraisals and the role of a given component are partly dependent on the current state of the environment.

2.3.4 Emotions Differ between Individuals. Besides the influence of situational context, there are individual differences in the appraisal of emotion-triggering events. The same situation can be appraised differently by different people, leading to different emotions [92]. Differences in personality (such as high versus low risk takers), affective states (such as moods or attitudes), and even biological factors (such as sex) influence emotion processing on a neural and psychological level [53]. Additionally, emotion regulation strategies differ between individuals, leading to varying changes in emotional expressions due to regulations before, during, and after the experience of emotions [51]. In other words, appraisals and the role of a given component are partly dependent on individual traits.

2.3.5 Emotions Evolve Over Time. Emotions are dynamic processes. For example, a threatening situation can evolve from anxious anticipation, to fear of a direct threat, to recovery from that stressor [141]. Some emotional experiences linger while others are fleeting, and the transition between emotions is not always discrete, allowing for mixing of different emotional states [6]. Important individual differences further affect the degree to which one can “move on” from a given emotional state to another [58, 106]. With all this in mind, emotional appraisals and the role of a given component are partly dependent on the progression of emotional experiences, the context, and individual differences.

2.3.6 Conclusion. In summary, building a theoretical basis for the model of player-game interactions on shared assumptions such as these provides the possibility to design for emotions without limiting the approach to one highly specific emotional perspective, which makes such a model flexible for different views and robust to changes in our understanding of emotions. Framing the game-player relationship in such a way allows developers to focus not only on established theoretical assumptions but also on specific perspectives with all their implications. Either way, such an approach could reduce the theoretical confusion that arises through inconsistent or poorly justified applications of theoretical assumptions.

3 DESIGN FRAMEWORKS BASED ON EMOTIONS

3.1 Design Research

For a long time, design research has been concerned with the emotional relationship between a product and a user and introduced many frameworks that model this relationship. Beginning in the 1990s, there was a high interest, both in industry (such as the technology industry) and in customers, to make products more engaging, more authentic, and easier to use by building designs based on key consumer emotions [32]. In the following years, Desmet [31] found no convincing relationship between a product and an emotion but rather a relationship between a product and

the emotional reaction of a specific person in a specific context. They concluded that affective responses resulted from appraised values that were complex, personal, and dependent on contextual factors such as time or environment. Exploring the impact of appraisal deeper, Norman [97] conceptualized three levels of design, based on cognitive processing: the visceral level for intuitive responses, the behavior level for consumption actions, and the reflective level of conscious experiences.

The idea that products can be designed from the perspective of being an experience for users established itself as a key school of thought in areas like HCI [86]. For example, Sonderegger and Sauer [130] found that aesthetic appeal of a phone application as appraised by users had an influence on usability ratings. The process of designing for emotional responses became known as Emotional Design and was communicated via channels like style, function, form, and usability based on users' needs and demands [59]. A key aspect of this process is user-centered design, meaning a clear understanding of the target user requirements [1]. Methods like interviews, questionnaires, focus groups, observations, and simulations were recommended for both understanding users in the beginning of a design cycle and evaluating the design at the final stages.

These principles were applied by different studies to showcase how a design could benefit from focusing on the emotional value. For example, Desmet et al. [33] showed that a "wow-effect" could be elicited in the design of a cell phone in three steps: first, a usable emotional conceptualization of the wow-effect; second, the combination of qualitative and quantitative research to provide rich knowledge of products that elicit similar experiences; and third, the focus on the product's character instead of the features to form a deeper relationship. Similarly, there are various examples where emotional design was applied to be appealing first, functional second, and finally create a relationship with the user for higher emotional impact [84, 99], often mirroring the three design levels of Norman [97].

3.2 Game Design

3.2.1 Experience Design. Naturally, the interest in emotions has been represented in the game industry since the very beginning, as games were considered entertainment products with the main purpose being the creation of positive experiences [72]. Salen et al.'s influential book *Rules of Play: Game Design Fundamentals* [114] discusses various game design concepts (such as rules, play, and culture) that are used to address second-order design problems, i.e., the indirect design of player experiences by directly designing game rules and environments. Adding to the notion that PXs are the focal point of game design, Schell [117] argued that the whole process of game design should be built around the elicitation of target experiences. This notion has been established in the game industry [52], leading to the creation of many works about emotional impact in games. For example, Isbister [67] described unique game qualities found in commercial games that hold emotional potential, such as choice, which is often associated with player agency and described as a way to create meaningful emotional reactions [133], or flow, which describes a state of concentration and deep, immersive enjoyment, as originally described by Csikszentmihalyi [26]. Chen [21] argued that each game can evoke an optimal flow state by balancing challenge and player abilities, which would maximize engagement and enjoyment for each game. In design practice, there are also many examples that utilize an extensive examination and re-creation of other successful examples in different contexts [117]. Such a practice is very much in line with media research work concerning users' draw toward certain mediums. The well-known Uses and Gratifications theory poses that users develop an understanding of the gratifications (for example, of an emotional nature) of certain media which is actively sought out [138]. In a game design context, this means that specific game aspects can be developed based on the expected emotional rewards of target audiences, potentially identified through successful examples of similar experiences [78].

In the following years, many frameworks have been developed to describe game elements that can have an effect on player emotions. For example, some studies provide adaptations of Norman's [97] three design levels specifically for game design [7, 28], arguing that games should make specific emotion elicitation decisions on a visceral, behavioral, and reflective level. The **Mechanics, Dynamics, and Aesthetics (MDA)** framework [65] is a popular game design framework that focuses on aesthetic game content as a central design component to elicit emotions and proposes an iterative approach to design for the aesthetic aspects of games. It also provides taxonomies of game elements that are potentially related to emotional reactions (such as narrative and challenge) and model how they relate to game systems. Similarly, the **Mechanics, Dynamics, Emotions (MDE)** framework [109] builds on the MDA to propose a model of gamification design. The emotion component here describes a player's state of mind and is used to explain the importance of emotions for a game experience in relationship to game dynamics and mechanics. Importantly, these frameworks acknowledge the value of emotions as a crucial design component within a game, without providing a deeper design guide or precise definition in a psychological sense.

3.2.2 Affective Design. In the last few years, affective computing has gained more attention as a way to provide insights about the affective game-player relationship. As a discipline within HCI, it is concerned with the measurement of and reaction to user emotions by a computer system [102]. It therefore represents an interactional relationship that has the potential to provide optimal emotional experiences by taking the current affective state of the user into account. In an effort to bring affective computing research to games, Hudlicka [62, 63] described three main components that are characteristic of affective games: emotion sensing and recognition, computational models of emotion, and emotion expression or adaptation.

There are unique challenges related to all of these components. For example, there is no universally accurate instrument to measure emotions, and recognition methods depend on emotion model assumptions. Similarly, computational models often make simplified assumptions about the nature of emotions, yielding little benefit to the user, while still being more difficult and expensive to develop than conventional games [54, 64]. Finally, adaptation comes with many challenges regarding adaptation rules and the mapping between measures and changes in the game [146]. Overall, technical limitations, theoretical uncertainties, and little representation in commercial games currently limit the use of affective design techniques in creating emotional games.

Still, affective design studies provide a shared high-level concept of the emotional interaction between player and game, known as the affective feedback loop [131]. The concept of the affective feedback loop originally involved embodied player emotions and adaptive game systems, reacting to changes in the other component respectively. It has since evolved, focusing not just on affective information measured through bodily responses but also on general affective information [61]. Because of this, there currently exist many versions of the affective feedback loop, including purely physiology-based technical implementations of affect-adaptive games [12], categorization tools for physiological game literature [108], or high-level conceptual models describing affective game system tasks [146]. Still, in all versions of the affective loop the closed nature is emphasized, as the ongoing adaptation of the game system to the changes in players' emotions is argued to be a unique characteristic of games compared to other mediums [100].

4 THEORETICAL MODEL OF THE EMOTIONAL GAME-PLAYER INTERACTION

As illustrated, a lot of progress has been made to frame the emotional relationship between players and video games. There is no doubt that many commercial video games do in fact have a considerable emotional impact and affective games have been popular and promising topics in game research. There are, however, some unclear concepts and inconsistent definitions within the whole area, leading to potential confusion about the emotional relationship between game and player.

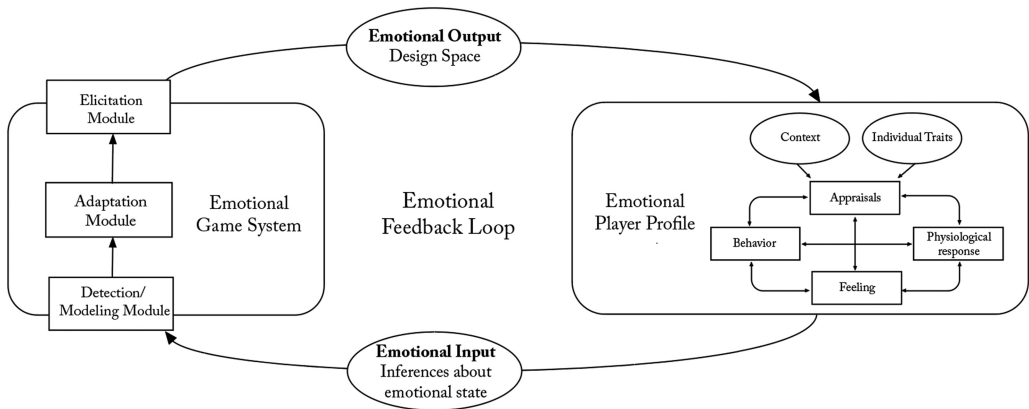


Fig. 3. Illustration of the emotional feedback loop as an expression of game-player interaction. The game system uses emotional input that infers information about player emotions and adapts the emotional output accordingly. The player appraises the output on multiple variables, which leads to interactions on measurable emotion components that can in turn be used to close the feedback loop.

Here we describe a theoretical model with clear definitions for each component and therefore integrate contemporary research from psychology, game design, and affective computing into one interactional model. This model expands the version of the affective feedback loop as described by Yannakakis and Paiva [146], specifically by implementing an emotion-theoretical model of the player into the loop and providing new labels to more precisely talk about emotional information that is being exchanged within the game-player interaction.

4.1 The Player

The new player component in the context of emotional games specifically means the player's emotions. Depending on the design perspective, emotions are sometimes defined as experiences, sometimes as action motivations, and sometimes as bodily reactions. Drawing from shared assumptions in emotion research, we can assume that (1) emotions consist of multiple components (like physiological, behavioral, and subjective feelings); (2) appraisal is central to the experience; (3) appraisal and the experience are dependent on context; (4) appraisal and the experience are dependent on individual differences; and (f) emotions evolve over time.

With this definition as a base, we are able to illustrate the player as part of the ongoing feedback loop with the game, adding specific emotions as psychological targets to the affective feedback loop that already exists in HCI studies [131, 146]. We focus here on emotions as short and intense affective states and aim to reduce the inconsistent and imprecise use of the word "affect," which is especially important given the variety of affective feedback loops present in the literature. This model therefore represents an *emotional* feedback loop (Figure 3). To be precise, it explains relevant concepts of the affective relationship between player and games in the domain of emotions and not other affective states, such as moods or traits.

4.2 The Game

In the discipline of affective games, a game system has per definition three major tasks it has to fulfill to become affective: emotion sensing and recognition, computational models of emotion, and emotion expression or adaptation [62]. Based on these tasks, Yannakakis and Paiva [146] provided descriptions of three game system modules: an emotion detection module (a module to measure and model player emotions), an adaptation module (a module to adapt the game world to these

player emotions), and an elicitation module (a module to elicit target player emotions). These modules represent the game side of the emotional feedback loop.

While these modules might not be explicitly described in traditional games, they can be used to describe familiar functions of any game system that relate to player emotions. For example, if a game offers a hard challenge, it has the potential to elicit feelings of satisfaction, accomplishment, and pride (emotion elicitation). There is, however, still the risk to evoke unwanted frustrating experiences; therefore, most games gradually increase the challenge based on prior performance or player choice (emotion adaptation). While this adaptation is not based on primary emotional data, it could be based on, for example, difficulty settings, i.e., subjective preference (emotion detection). Every game that builds an emotional relationship can therefore be viewed within the emotional feedback loop.

4.3 Closing the Loop

Given the game system and the emotional player profile, there are now two kinds of information that are being exchanged. First, there are the stimuli provided by the game for the player. This is the design space of the game, which includes possible stimuli that elicit certain emotions. We propose the term *emotional output* for these stimuli.

We choose this specific concept to describe game aspects that have a relationship to player emotions. This relationship cannot be seen as universal, which means it is dependent on individual differences and context. Following the loop, the player expresses the emotional reaction in various components (such as feeling, physiological response, or behavior). This expression includes information provided by the player to be measured through the game to infer emotional states. We conceptualize this information as *emotional input*. Naturally, emotional input can consist of direct emotion measures, like in biofeedback games. However, physiological measures cannot be assumed to map to certain emotions [85], and given the complex nature of emotions and the varying experiences games want to achieve, input measures need to be considered in terms of accuracy, cost, effort, and usefulness [35]. Therefore, other information might be more appropriate to use, like indirect measures, which could include any input a player makes in a game, from button-press patterns to in-game behaviors. Furthermore, building on our knowledge of how emotions occur, inferences could also be made from assumptions about contextual aspects, individual player traits, and emotion progressions that relate to the target emotion.

4.4 Summary

To summarize, the emotional feedback loop provides terms to describe the emotional game system (and its tasks), the emotional player profile with relevant emotion-theoretical components, and the information that is being exchanged between the game and the player. We propose this model as an extension of the affective feedback loop [131, 146] to provide (1) clear terms that are consistent with their definitions in their respective fields and (2) a comprehensive illustration with integrated, contemporary knowledge regarding emotion research, game design, and affective computing. This model can be used to more easily talk about aspects of the emotional game-player interaction. It is therefore useful in terms of describing functional designs for existing games, guidelines for developing emotional games (based on the knowledge we currently have), and researching emotional phenomena that are part of such an interaction.

For example, using this model, we can summarize the tasks of designing emotional games as follows:

- (1) The **design of emotional game output**: The creation of game stimuli that elicit a certain target emotion through appraisal, given a certain context and individual differences.

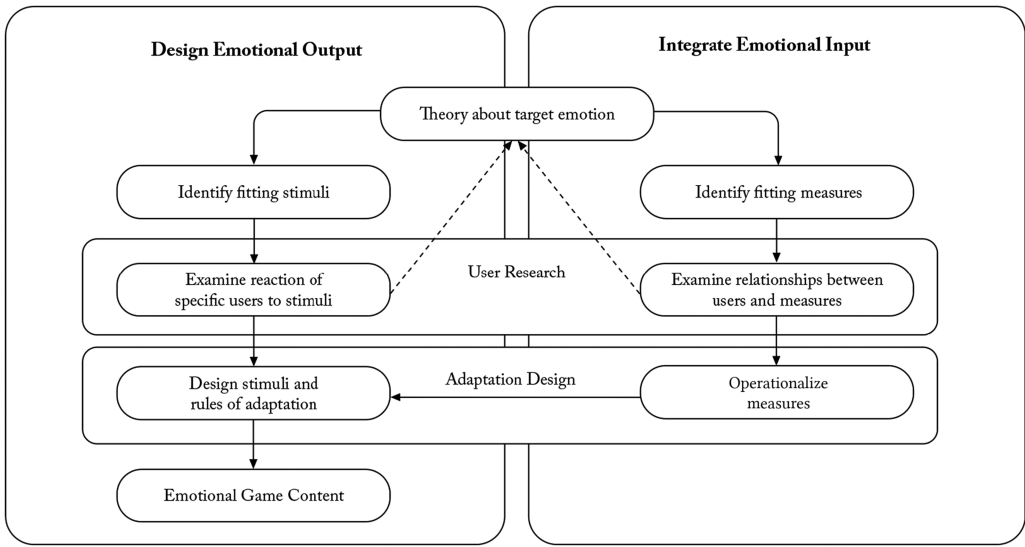


Fig. 4. Illustration of the practical framework: To elicit a target emotion, both the domain of the game material and the domain of the user input needs to be considered. Potential game characteristics (domain 1) and user input variables (domain 2) that could relate to the target experience need to be identified (step 1) and tested in user research (step 2). Gathered information can be used to introduce adaptivity of the game material based on user input to bring the product closer to the intended experience (step 3) or to improve research by further developing theoretical assumptions about the target emotion (dashed arrows).

- (2) The **integration of emotional game input**: The measuring of certain emotion expressions from the player that can be used to adapt the game in favor of the target emotion.

In order to contribute beyond research integration and new terminology, we now use the descriptive tools provided by the model to propose a framework for designing and researching emotional games.

5 PRACTICAL FRAMEWORK FOR EMOTION DESIGN FOR VIDEO GAMES

The aim of this section is to provide a practical guide for game designers and researchers to develop and test video games or video game characteristics that are adaptive to player emotions. It builds upon the emotional feedback loop to address some of the issues related to emotion elicitation in games and streamlines the process of integrating emotional adaptivity into game projects. Building on other interactional design frameworks [59], this process will be referred to as **emotion design for video games**. An illustration of the process can be viewed in Figure 4. To provide the most useful overview of possible methods to apply the emotional feedback loop, it is important to further avoid confusion with other subtly different concepts and accurately describe the individual process steps.

5.1 Theory about Target Emotion

In order to design a game or game aspect that elicits a certain emotion, the target emotion in question needs to be conceptualized. In game design practice, the affective reward of games is often referred to as the experience [117], which can be considered as the subjective component of a certain emotional reaction. For example, the emotion “sadness” includes as previously discussed many different components (e.g., of a cognitive, physiological, and behavioral nature), but the

target itself for the designer is the subjective, conscious feeling of sadness. When choosing a target emotion, we can therefore at this stage consider only the subjective feeling a designer wants to elicit in a player.

Naturally, there is a growing interest in experiences that relate to not only broad “basic” emotions (e.g., fear or happiness) but also more complex and underexplored emotions [30], including “aesthetic” emotions (e.g., *schadenfreude* or the feeling of being moved [13]), or even emotions that have no concrete label (e.g., the anticipation of something bad happening or ordinary, less intense emotions [137]). Considering that the labels of emotions also often differ between languages and cultures [39], the choice of target emotion should not be limited to well-known labels but might also include experiences that are not yet fully understood or defined. Successful emotion elicitation might be facilitated if the target emotion has already been thoroughly explored in psychological research, but the only necessary characteristic of the target emotion is that it can be evaluated in terms of how successful the elicitation is.

When a target emotion is chosen, the subsequent process can be considered from two domains. The domain of emotional output tries to identify game material that evokes the target experience. The question that guides the steps in this domain is: what game characteristics evoke the target emotion? The domain of emotional input, on the other hand, tries to identify measures a game can use as indication of the player emotion. The question that guides the next steps in this domain is: what can a game track that indicates the target emotion? The framework introduces the terminology and process steps for these two domains for two reasons: (1) both domains need to be considered to ensure the target emotion can be elicited across individuals, and (2) both domains need to be considered to ensure the target emotion is elicited for one player during the progression of the game.

The framework specifically addresses uncertainties regarding emotion expression, differences in emotion elicitation depending on individuals and context, and the complex progression of emotions.

5.2 First Domain: Design Emotional Output

5.2.1 Identify Fitting Stimuli. The literature describes many possible game elements that evoke certain emotional reactions, for example, NPCs [81], avatars [18], game world, mechanics and narrative [146], and graphics and music [74], but also more high-level concepts like rewards, creativity, and player agency [67, 94]. Frameworks like *emotioneering* [40] or the MDE framework [109] provide a variety of possible channels to use in the game design process.

It is, however, crucial to acknowledge that the characteristics of a stimulus are only important in the context of appraisal: it is not a fixed attribute of the dark hallway to evoke fear. Considering the classic appraisal variables described by Lazarus [77], goal-relevant but incongruent stimuli that are difficult to cope with correspond to fear, so the appraisal of the dark hallway as scary is dependent on specific features: Does the player need to follow the hallway to reach a goal? Do players feel like the darkness is incongruent to their goal?

At this stage, game materials can be designed based on assumptions or existing examples indicating their potential in eliciting the target emotion. While it can be helpful to review known appraisal variables for well-researched emotions, the more practical and common approach is to design based on successful examples. For example, in an extensive analysis of the emotional effects of *Silent Hill* by Perron [101], the author discusses how the specific type of emotional terror in the *Silent Hill* series was designed to mimic Japanese psychological horror movies and how it differs from the experiences of other horror media. This discussion is led by appraisal on a reflective processing level. For example, during the production of *Silent Hill 2*, the developers tried to induce anxiety gradually, increasing conscious anticipation of negative events [101]. Other processing

levels can be considered, including sensorimotor responses (e.g., fear through sounds [50]) and perceptual and associative responses (e.g., fear through danger-associated stimuli).

5.2.2 Examine Reaction of Specific Users to Stimuli. Studies about industry practices show that game companies often rely on genre conventions, historical successes of other games, or internal theories of the developers when designing game stimuli [30, 52], and to great success. As observed through the lens of Uses and Gratification theory, players too make use of genres, their knowledge of other games, and their own internal theories to predict their own feelings when playing a specific game [138].

But to make sure the game (or game elements) really evoke the target experience, game creators need to test them, and this is commonly done through user research to thoroughly understand the individual emotional reactions of users within a target audience. In a 2018 discussion of game user research [148], Zammito of Electronic Arts states that almost all large game companies have teams of dedicated user researchers that are responsible for executing UX tests of a qualitative or quantitative nature to inform developers of the experiences of target audience members during various cycles of game development. Through organized processes involving design, development, and testing, many game aspects are iteratively fine-tuned to deliver on the target experience [34]. Within the domain of emotional output, user research can establish whether a game aspect can elicit the target emotion for certain individuals and in certain contexts.

The methods to conduct such research are often discussed in literature about player research or targeted empirical tests and usually include self-report instruments, such as questionnaires (e.g., the PANAS [140]), where players rate how strongly they felt an emotion in a certain moment of the game after playing. This provides a precise quantitative measure of experiences that is easy to interpret, but may be influenced by restrictions of the questionnaire (e.g., not the complete emotional space is included). To capture a more complex view of the experiences, interviews or open questions can be constructed where players describe the experience in their own words (see, for example, [1]). This has the potential to provide a more detailed overview of the elicited emotions but might be harder to interpret for the use within a game. Additionally, self-reports after game-play sessions could be influenced by factors like memory effects and only capture the most notable emotional reactions. To account for the evolving nature of emotions with variable onset and offset periods and subtle differences over time, continuous experience reports could be used (such as with the two-dimensional emotion-space [121], the affect rating dial [110], or even free association techniques [60]). Players get the opportunity to report their experiences continuously during gameplay and data can be collected that map exactly to in-game moments. Again, responses are limited to how open the scale is, and free responses are more difficult to interpret and quantify. There is an exceedingly large space to describe experiences both in categories and in finer gradients [23], so the choice of research methods should account for the needs of the specific designs.

5.2.3 Design Stimuli and Rules of Adaptation. As mentioned before, games and specific game aspects are most commonly designed through iterations that are being evaluated in their potential to elicit a target experience. Using the proposed terminology of the framework, we can say that game stimuli are evaluated in their potential to become emotional output. This implicitly means for the designer that there is an at least partly known relationship between the game they create and the emotion a user experiences when playing it. Broadly speaking, this can mean that a game consists of similar elements that have been successful in other works, delivering an experience players can expect and seek out, but it can also mean on a detail level that specific aspects of a game have been extensively tuned and tested to ensure a certain emotional experience, even when played by a variety of players.

This by itself is not a new process, but we argue that given the complexity of emotion elicitation and expression, two criteria for the success of creating emotional game material should be considered at this stage: (1) how well does this design work across individuals and (2) how well does this design work given the rapidly changing emotions within one individual.

We call the process that aims to ensure that these questions are addressed adaptation. Interindividual adaptation attempts to create the same emotional experience for different kinds of players. For example, players with a high ability to play a certain genre may enjoy more challenging gameplay, while players with lower ability would experience the same enjoyment with less challenge. Ideally, user research has identified game characteristics (such as reaction time windows or health for challenge) that are the source of different experiences for different types of players, so these characteristics can be accounted for (e.g., through purposeful designed difficulty settings).

Intraindividual adaptation attempts to create an optimal experience for one player as the game unfolds, based on assumptions of how emotions evolve. For example, a game has successfully elicited fear for a player, which means that the design space is now open for more intense fear reactions (e.g., terror) or strong opposite emotions (e.g., the feeling of relief).

If a design can elicit the target emotion across different individuals and within one individual, it can be considered adapted to player emotions and therefore as emotional output. We can illustrate this with the example of game difficulty: it is widely acknowledged that game difficulty has an influence on enjoyment when player abilities are taken into account (e.g., in dynamic difficulty adjustments or in player-chosen difficulty settings) [3]. Many games provide difficulty options to ensure enjoyable experiences for different types of players but also adapt game difficulty during the course of the game to accommodate the changing abilities of one player. In order for this to work, game designers need to be aware of the relationship between difficulty settings and enjoyment, i.e., emotional output and target emotion, as discussed in the previous sections. But they also need to be aware of the current state of the player, which the framework addresses in the second domain. In this specific example, we need some kind of input from the player to determine if they are enjoying the game, either as measures of their experience [3] or their selection of difficulty settings. This input that infers the emotional state of the player and can be used for adaptation and successful emotion elicitation is referred to as the emotional input.

5.3 Second Domain: Integrate Emotional Input

5.3.1 Identify Fitting Measures. Emotional input in a game design context could be any data that holds information about the emotional profile of the player in the current moment of the game-player interaction. Within the domain of emotional input, the designer tries to identify player input variables that are related to the target emotion so that these can be used for game adaptation.

In affective gaming, there are many other potential input variables measured that could be useful (depending on the game). The most prevalent variables are of a physiological nature [108], and multiple instruments are commonly investigated to assess them: **electrocardiography (ECG)** measures heart activity, which can include both **heart rate (HR)** and **heart rate variability (HRV)**. In gaming contexts, HR and HRV are often related to stress or fear [12, 29, 96, 139]. **Electrodermal activity (EDA)** or **Galvanic Skin Response (GSR)** refers to the skin electrical conductivity, which can be measured through sensors that are small, wearable, and cheap. It is commonly used to reflect arousal aspects of emotions, which is used in some affective gaming studies to infer information about high-arousal emotions, such as fear or excitement [22, 29, 96]. **Electromyography (EMG)** measures the electrical potential of muscle cells, which is commonly used in gaming contexts to make inferences about valence aspects of emotions by measuring facial muscle activities [96, 134]. Finally, **electroencephalography (EEG)** is used to measure the electric activity of the brain, which has been found to be closely related to reports of valence and arousal, but

includes multiple sensors and complex data that are sensitive to any changes in brain activity [35]. Whether or not specific emotions map directly onto specific physiological profiles across all contexts remains doubtful [85], so the validity and reliability of each instrument need to be tested given a game context and specific target emotion. Nevertheless, physiological sensors offer a range of primary affective data that can be used to test assumptions about their relationships with target emotional experiences [71, 85].

Another class of measures considers observational behaviors, such as gestures, postures, facial expressions, or eye gaze. Motion capture has had some success in classifying emotions in regard to either high or low intensity [116]. Measurement of facial expressions has been used to distinguish between emotional facial features and is reported to be easy and unobtrusive to implement [10]. However, accuracy seems to depend on players being in the right position, being similarly expressive, and not manipulating reactions [12]. Some emotions can also be expressed in multiple ways, which could limit the practicability of such an approach. Another measure lies in eye gaze, which is typically associated with attention, but has also been used to provide simple classification information on arousal or valence aspects of emotions [82, 129] or to create new gameplay input systems [93]. While eye gaze is hard to track in classical gaming setups, virtual reality headsets with included eye-tracking devices are becoming more popular, which makes this technology for some games easy to implement and a promising source of data for adaptation [103, 128].

Finally, in-game behavior can be used to infer information about current emotions as players have the opportunity to influence the game through their input (or player choice [67]). For example, input behavior in mobile games has been shown to predict low and high levels of valence and arousal [46], and certain emotional states are related to input variables like touch pressure [90] or controller button presses [132]. There have been efforts made to model emotional states from in-game performance or input parameters (e.g., [44, 123]), and most recently, emotion classification based on in-game responses to narratives has been used to measure distinct emotional states [70]. This is still a young field, but game adaptation based on behavior yields promising results in engaging different types of players [16, 136].

In summary, there are many possible sources of information, although none are universally applicable. The choice depends on the design aim and requirements but should in every case be informed by user research.

5.3.2 Examine Relationships between Users and Measures. As with the relationship between game material and emotional reaction, we often cannot assume that the relationship between player input and their current emotional state is known. Although there is a wide variety of measures that have been related to emotion in one way or another, there is no perfect solution to measure emotions in games. Robust, universal one-to-one matchings between emotions and measurement either have yet to be discovered or do not exist. Some researchers in the field argue for a combination of different methods, which seems necessary to increase accuracy but also increases cost and effort.

Designers can, however, avoid these uncertainties by focusing on the project's specific design aims. Since the resulting subjective experience is the design aim, games do not need to accurately predict all emotions, but rather explore the relationship between appropriate available data and the subjective experience of the target emotion. When testing the effects of the games on users' emotion through user research (as explained in Section 5.2.2), games can also test what kinds of user input can be used to assess the current player emotion.

This means that the given data should provide some indication of the relative presence or absence of the targeted emotional state within the tested game context and for the tested audience. However, it need not distinguish between that state and other unlikely states. A game that aims

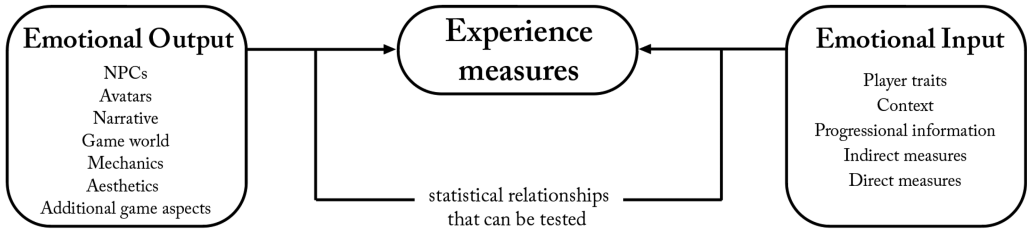


Fig. 5. Model of relevant variables for the user research that allows the connection of emotional output and emotional input through experience measures given a game context. The tested statistical relationships can be used to inform the design of emotional evocative games, without the need for specified emotion models.

at relaxation does not need to rely on accurately measuring sadness. However, it should establish some kind of indication of subjective relaxation.

In order to be considered adaptive the game must demonstrate an effect on player relaxation for different individuals and during different phases of the game. The main question to address in the user research step for the second domain is therefore: what information can a game use to identify such differences and adapt? For a physiological relaxation game that uses HR measures, user research can explicitly test the relationship between HR and relaxation for different individuals. If it exists in the context of the game and is usable, the game has now a way to track the target emotion of the player and adapt to ensure it, regardless of the individual differences or the current point of the game. For another relaxation game without physiology tracking, user research can, for example, explicitly test the relationship between in-game performance and relaxation and, if this proves useful, can then design difficulty levels that ensure a specific in-game performance and therefore relaxation.

Like in the first domain, measures can become usable to infer emotional states by specifically testing their statistical relationships within the context of the game. An illustration of the user research for both domains of the framework is depicted in Figure 5.

5.3.3 Operationalize Measures. User research informs about the data sources that can be used to make inferences about emotional states, so the last remaining step is to use the gathered knowledge to connect both domains of the framework, as depicted in Figure 5. Both the emotional input and the emotional output of a game have an established relationship with the target experiences, so the input can be used to determine the output.

An illustration of such a process can be seen in the study by Shaker et al. [124], who collected gameplay behavior and self-reported experiences from players first and then used the findings to identify important input patterns that reflected frustration, challenge, and fun. Game worlds were then created that adapted to these input patterns, which resulted in a better evaluation of the game. Consequent studies showed that collecting gameplay data beforehand provided the opportunity to model player experiences, which could point to the practicability of such a process for almost all games [123].

In this sense, the emotional input and the emotional output are both representations of the experience and are therefore easily exchangeable and enhanceable in reaction to the needs, technical progress, and more informed theories. Such a modularization is currently heavily emphasized by researchers in affective computing, not only because it can make actual solutions easier and more approachable, but also because theories from psychology, design, and HCI are more and more in the process of unification and a system that draws on and informs all of these fields will need to be dynamic [107]. For example, if face recognition is used as a measure for anxiety within a game,

it can be complemented by other measures or even easily exchanged, because it only represents the tested relationship with the experience and not an underlying model of emotion that makes face recognition necessary.

5.4 Summary

To ensure that a game elicits a target experience, this framework provides the terminology and methodology to make some of the unknowns explicitly known and usable to design game elements. This process includes:

First Domain: Design Emotional Output

- (1) **Identify fitting stimuli.** Make use of common design practices (e.g., the use of successful examples, game design frameworks) to identify game elements with the potential to elicit the target experience.
- (2) **Examine reaction of specific users to stimuli.** Explicitly test the relationship between the designed game characteristic and the target emotion using user research methods. This step needs to take individual differences and game progression into account.
- (3) **Design stimuli and rules of adaptation.** Given the knowledge of step 2, determine how the game needs to adapt between individuals and during the progression of the game to ensure the target experience. This step needs to take the second domain into account.

Second Domain: Integrate Emotional Input

- (1) **Identify fitting measures.** Identify ways to indicate whether or not the target emotion is experienced with consideration to cost, invasiveness, generalizability, and ease of interpretation.
- (2) **Examine relationship between users and measures.** Explicitly test the relationship between the chosen measures and the target emotion using user research methods.
- (3) **Operationalize measures.** Given the knowledge of step 2, determine how user input can be used to trigger the adaptations of the first domain.

This framework was developed to help avoid outdated classification techniques or simplified theoretical assumptions. To bring video games closer to the benefits associated with adaptation, this process can be followed for certain games or game aspects, while issues currently present in affective games research (as such rules that are unlikely to hold across all situations and individuals) are being addressed. Designers and developers should be able to test relationships themselves for a specific game context and player group to make informed decisions about how to adapt a game to player emotions. This is necessary because there are many unanswered questions about the nature, variety, and underlying mechanisms of emotion. This framework does not claim to provide answers to these questions; instead, the aim is to illustrate methods that could help designers and researchers get closer to finding some of the answers themselves. In fact, because we are far from a common understanding about how emotions can be modeled, it is more useful to enable developers and researchers to contribute without the need for a comprehensive understanding of emotions.

5.5 Limitations of the Framework

The process presented here showcases the necessary steps to create adaptivity to player emotions. In essence, there are two domains of information that need to be considered, namely the relationship between the game itself and the target emotion, and the relationship between player input and the target emotion. Both these domains are necessary for adaptivity, but contemporary research in psychology, design, and affective gaming shows that there are very little assumptions regarding these domains that hold true across populations and contexts. As a consequence,

user tests become an integral part of fundamental design decisions, and they come with considerable cost. Depending on the target emotion, there are few validated measurement instruments, so exploratory fundamental research is necessary to even find ways to detect the target emotion, which might not be practical in a production environment. And depending on the differences between players, some games might need to implement very complex player models that are costly to develop and run and might have a limited gain [20]. Still, as user research is an important part of many game projects already [148], the additional resources necessary to create a more emotionally adapted game compared to potential gains [146] really depend on the project and the aims. While cost could exponentially grow when adapting every single game element to every possible player and situation, there are many examples of the benefits for some kind of emotion adaptivity, ranging from broad game aspects [12] to complex procedurally generated game worlds [147]. Adaptivity can bring a game closer to the intended experience but can become very costly for high levels of personalization. Since humans are not purely reactive but active participants in constructing their experiences [8], we argue that the main potential of the process lies not within completely personalized experiences for every player in every situation, but in specific gaps between intended experience and observed experience. As of now, there is more work necessary to gauge how costs and benefits balance for different types of projects and levels of adaptation.

6 CONCLUSION

This article provides an overview of the literature regarding psychological models of emotion, design research, and HCI research with a focus on affective computing. Common and most recent findings were combined to provide a model of the ongoing emotional interaction between a player and a game that can be represented by our refined version of the emotional feedback loop (Figure 3). Based on this loop, a methodological framework of emotion design for video games was illustrated that introduced explicit methods to implement emotion adaptivity in games and to research some of the open questions that remain about our emotional connection to games. As Caroux [17] reported, there is a lack of unity in the work that conceptualizes game-player interactions. Our approach takes a step toward reducing this lack. It connects theoretical perspectives from multiple disciplines. The proposed approach was created to reduce confusion and to provide guidance for applied emotion research games that does not restrict them by oversimplifying their assumptions about emotions or misinterpreting relationships between affective variables. It is crucial to not expand the confusion even more, but to connect different fields with a common theory that is both dynamic and useful. Doing this, a design-led approach could address some of the technological and conceptual problems that currently exist in the affective game literature. To achieve this, it is necessary to examine how this framework could be integrated with common design practice in a way that supports the process. A recent manuscript providing a large-scale evaluation study applied the framework to the development of an affective 2D arcade game and conducted a randomized controlled trial to study the effects of an affect-adaptive version of the game in comparison to a control version with results in terms of subjective valence increase [24]. To further the usefulness of this contribution, more applications need to be studied.

An emotional game is not (yet) a game that completely understands the full range of human emotions, but a game that can successfully create an emotional experience. Only with an approach that is robust to uncertainties but still enabling in the questions that matter can designers and researchers understand more about the processes that are involved.

7 DISCLOSURE STATEMENT

The authors report there are no competing interests to declare.

REFERENCES

- [1] Chadia Abras, Diane Maloney-Krichmar, Jenny Preece, et al. 2004. User-centered design. In *Encyclopedia of Human-computer Interaction*, W. Bainbridge (Ed.). Sage Publications, Thousand Oaks, CA. 445–456.
- [2] Hussein Al Osman, Haiwei Dong, and Abdulmotaleb El Saddik. 2016. Ubiquitous biofeedback serious game for stress management. *IEEE Access* 4 (2016), 1274–1286.
- [3] Justin T. Alexander, John Sear, and Andreas Oikonomou. 2013. An investigation of the effects of game difficulty on player enjoyment. *Entertainment Computing* 4, 1 (2013), 53–62.
- [4] Luigi Anolli, Fabrizia Mantovani, Linda Confalonieri, Antonio Ascolese, and L. Peveri. 2010. Emotions in serious games: From experience to assessment. *International Journal of Emerging Technologies in Learning (iJET)* 5, SI3 (2010), 7–16. <https://doi.org/10.3991/ijet.v5iSI3.1496>
- [5] Magda B. Arnold. 1960. *Emotion and Personality*. Columbia University Press.
- [6] Erkin Asutay, Alexander Genevsky, Lisa Feldman Barrett, J. Paul Hamilton, Paul Slovic, and Daniel Västfjäll. 2021. Affective calculus: The construction of affect through information integration over time. *Emotion* 21, 1 (2021), 159–174. <https://doi.org/10.1037/emo0000681>
- [7] Shahrel Nizar Baharom, Wee Hoe Tan, and Mohammad Zaffwan Idris. 2014. Emotional design for games: A framework for player-centric approach in the game design process. *International Journal of Multimedia and Ubiquitous Engineering* 9, 10 (2014), 387–398.
- [8] Lisa Feldman Barrett. 2017. *How Emotions Are Made: The Secret Life of the Brain*. Houghton Mifflin Harcourt.
- [9] Daniel Bersak, Gary McDarby, Ned Augenblick, Phil McDarby, Daragh McDonnell, Brian McDonald, and Rahul Karkun. 2001. Intelligent biofeedback using an immersive competitive environment. In *Designing Ubiquitous Computing Games Workshop at UbiComp*. Citeseer, 1–6.
- [10] Paris Mavromoustakos Blom, Sander Bakkes, Chek Tan, Shimon Whiteson, Diederik Roijers, Roberto Valenti, and Theo Gevers. 2014. Towards personalised gaming via facial expression recognition. In *Proceedings of the AAAI Conference on Artificial Intelligence and Interactive Digital Entertainment*, Vol. 10.
- [11] Kirsten Boehner, Rogério DePaula, Paul Dourish, and Phoebe Sengers. 2007. How emotion is made and measured. *International Journal of Human-Computer Studies* 65, 4 (2007), 275–291.
- [12] Boyan Bontchev. 2016. Adaptation in affective video games: A literature review. *Cybernetics and Information Technologies* 16, 3 (2016), 3–34.
- [13] Julia Ayumi Bopp, Elisa D. Mekler, and Klaus Opwis. 2016. Negative emotion, positive experience? Emotionally moving moments in digital games. In *Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems*. 2996–3006.
- [14] Martin Bruder, Dina Dosmukhambetova, Josef Nerb, and Antony S. R. Manstead. 2012. Emotional signals in nonverbal interaction: Dyadic facilitation and convergence in expressions, appraisals, and feelings. *Cognition & Emotion* 26, 3 (2012), 480–502.
- [15] John T. Cacioppo, Gary G. Berntson, Jeff T. Larsen, Kirsten M. Poehlmann, Tiffany A. Ito, et al. 2000. The psychophysiology of emotion. *Handbook of Emotions* 2 (2000), 173–191.
- [16] Alessandro Canossa. 2009. *Play-persona: Modeling Player Behavior in Computer Games*. Danmarks Designskole.
- [17] Loïc Caroux, Katherine Isbister, Ludovic Le Bigot, and Nicolas Vibert. 2015. Player–video game interaction: A systematic review of current concepts. *Computers in Human Behavior* 48 (2015), 366–381.
- [18] Mickaël Causse, Bernard Pavard, Jean-Michel Senard, Jean-François Démonet, and Josette Pastor. 2012. Positive and negative emotion induction through avatars and its impact on reasoning performance: Cardiovascular and pupillary correlates. *Studia Psychologica* 54 (2012), pp–37.
- [19] Henrik Cederholm, Olle Hilborn, Craig Lindley, Charlotte C. Sennersten, and Jeanette Eriksson. 2011. The aiming game: Using a game with biofeedback for training in emotion regulation. In *DiGRA Conference*.
- [20] Darryl Charles, Michael McNeill, Moira McAlister, Michaela Black, Adrian Moore, Karl Stringer, Julian Kücklich, and Aphra Kerr. 2005. Player-centred game design: Player modelling and adaptive digital games. In *Proceedings of DiGRA 2005 Conference: Changing Views – Worlds in Play*. Digital Games Research Association: DiGRA, 285–298.
- [21] Jenova Chen. 2007. Flow in games (and everything else). *Communications of the ACM* 50, 4 (2007), 31–34.
- [22] Thomas Christy and Ludmila I. Kuncheva. 2014. Technological advancements in affective gaming: A historical survey. *GSTF Journal on Computing* 3, 4 (2014), 32–41.
- [23] Alan S. Cowen and Dacher Keltner. 2017. Self-report captures 27 distinct categories of emotion bridged by continuous gradients. *Proceedings of the National Academy of Sciences* 114, 38 (2017), E7900–E7909.
- [24] Maximilian Croissant, Guy Schofield, Madeleine Frister, and Cade McCall. n.d. *Advancing Methodological Approaches in Affect-Adaptive Video Game Design: Empirical Validation of Emotion-Driven Gameplay Modification*. Vol. 37, Manuscript submitted for publication.
- [25] Maximilian Croissant, Guy Schofield, and Cade McCall. 2023. Theories, methodologies, and effects of affect-adaptive games: A systematic review. *Entertainment Computing* (2023), 100591. <https://doi.org/10.1016/j.entcom.2023.100591>

- [26] Mihaly Csikszentmihalyi and Isabella Selega Csikszentmihalyi. 1992. *Optimal Experience: Psychological Studies of Flow in Consciousness*. Cambridge University Press.
- [27] Charles Darwin and Phillip Prodger. 1998. *The Expression of the Emotions in Man and Animals*. Oxford University Press.
- [28] Penny De Byl. 2015. A conceptual affective design framework for the use of emotions in computer game design. *Cyberpsychology: Journal of Psychosocial Research on Cyberspace* 9, 3, Article 4 (2015). <https://doi.org/10.5817/CP2015-3-4>
- [29] Andrew Dekker and Erik Champion. 2007. Please biofeed the zombies: Enhancing the gameplay and display of a horror game using biofeedback. In *Proceedings of the 2007 DiGRA International Conference: Situated Play (DiGRA '07)*. 550–558.
- [30] Alena Denisova, Julia Ayumi Bopp, Thuy Duong Nguyen, and Elisa D. Mekler. 2021. Whatever the emotional experience, It's Up to Them: Insights from designers of emotionally impactful games. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems (CHI'21)*. Association for Computing Machinery, New York, NY, 1–9. <https://doi.org/10.1145/3411764.3445286>
- [31] Pieter Desmet. 1992. MA 2002. Designing Emotions (Doctoral dissertation, Doctoral Dissertation. TU Delft, Netherlands).
- [32] Pieter M. A. Desmet and Paul Hekkert. 2009. Special issue editorial: Design & emotion. *International Journal of Design* 3, 2 (2009), 1–6.
- [33] Pieter M. A. Desmet, Rick Porcelijn, and M. B. Van Dijk. 2007. Emotional design: Application of a research-based design approach. *Knowledge, Technology & Policy* 20, 3 (2007), 141.
- [34] Anders Drachen, Pejman Mirza-Babaei, and Lennart E. Nacke. 2018. *Games User Research*. Oxford University Press.
- [35] Andrius Dziedzickis, Artūras Kaklauskas, and Vytautas Bucinskas. 2020. Human emotion recognition: Review of sensors and methods. *Sensors* 20, 3 (2020), 592.
- [36] Maria Egger, Matthias Ley, and Sten Hanke. 2019. Emotion recognition from physiological signal analysis: A review. *Electronic Notes in Theoretical Computer Science* 343 (2019), 35–55.
- [37] Paul Ekman. 1992. An argument for basic emotions. *Cognition & Emotion* 6, 3–4 (1992), 169–200.
- [38] Paul Ekman. 1971. Universals and cultural differences in facial expressions of emotion. *Nebraska Symposium on Motivation* 19 (1977), 207–283.
- [39] Alan Page Fiske. 2020. The lexical fallacy in emotion research: Mistaking vernacular words for psychological entities. *Psychological Review* 127, 1 (2020), 95.
- [40] David Freeman. 2004. Creating emotion in games: The craft and art of emotioneering™. *Computers in Entertainment (CIE)* 2, 3 (2004), 15–15.
- [41] Nico H. Frijda. 1993. Moods, emotion episodes, and emotions. In *Handbook of Emotions*, M. Lewis and J. M. Haviland (Eds.). The Guilford Press, 381–403.
- [42] Nico H. Frijda. 2009. Emotions, individual differences and time course: Reflections. *Cognition and Emotion* 23, 7 (2009), 1444–1461.
- [43] Nico H. Frijda, Batja Mesquita, Joep Sonnemans, Stephanie Van Goozen, and K. T. Strongman. 1991. The duration of affective phenomena or emotions, sentiments and passions. *International Review of Studies on Emotion* 1 (1991), 187–225.
- [44] Julian Frommel, Claudia Schrader, and Michael Weber. 2018. Towards emotion-based adaptive games: Emotion recognition via input and performance features. In *Proceedings of the 2018 Annual Symposium on Computer-human Interaction in Play*. 173–185.
- [45] Rockstar Games. 2018. *Red Dead Redemption 2*. [Game].
- [46] Yuan Gao, Nadia Bianchi-Berthouze, and Hongying Meng. 2012. What does touch tell us about emotions in touchscreen-based gameplay? *ACM Transactions on Computer-human Interaction (TOCHI)* 19, 4 (2012), 1–30.
- [47] Kiel Gilleade and Alan Dix. 2004. Using frustration in the design of adaptive videogames. In *Proceedings of the 2004 ACM SIGCHI International Conference on Advances in Computer Entertainment Technology*. 228–232.
- [48] Kiel Gilleade, Alan Dix, and Jen Allanson. 2005. Affective videogames and modes of affective gaming: Assist me, challenge me, emote me. In *DiGRA 2005: Changing Views—Worlds in Play*.
- [49] Roger Giner-Sorolla. 2019. The past thirty years of emotion research: Appraisal and beyond. *Cognition and Emotion* 33, 1 (2019), 48–54.
- [50] Steve Goodman. 2012. *Sonic Warfare: Sound, Affect, and the Ecology of Fear*. Technologies of Lived Abstraction.
- [51] James J. Gross and Oliver P. John. 2003. Individual differences in two emotion regulation processes: Implications for affect, relationships, and well-being. *Journal of Personality and Social Psychology* 85, 2 (2003), 348.
- [52] Ulf Hagen. 2011. Designing for player experience: How professional game developers communicate design visions. *Journal of Gaming & Virtual Worlds* 3, 3 (2011), 259–275.

- [53] Stephan Hamann and Turhan Canli. 2004. Individual differences in emotion processing. *Current Opinion in Neurobiology* 14, 2 (2004), 233–238.
- [54] Salma Hamdy and David King. 2017. Affect and believability in game characters—A review of the use of affective computing in games. In *Proceedings of the 18th Annual Conference on Simulation and AI in Computer Games*.
- [55] Eddie Harmon-Jones. 2019. On motivational influences, moving beyond valence, and integrating dimensional and discrete views of emotion. *Cognition and Emotion* 33, 1 (2019), 101–108.
- [56] Eddie Harmon-Jones, Cindy Harmon-Jones, and Elizabeth Summerell. 2017. On the importance of both dimensional and discrete models of emotion. *Behavioral Sciences* 7, 4 (2017), 66.
- [57] Olle Hilborn, Henrik Cederholm, Jeanette Eriksson, and Craig Lindley. 2013. A biofeedback game for training arousal regulation during a stressful task: The space investor. In *International Conference on Human-computer Interaction*. Springer, 403–410.
- [58] Lea K. Hildebrandt, Cade McCall, Haakon G. Engen, and Tania Singer. 2016. Cognitive flexibility, heart rate variability, and resilience predict fine-grained regulation of arousal during prolonged threat. *Psychophysiology* 53, 6 (2016), 880–890.
- [59] Amic G. Ho and Kin Wai Michael G. Siu. 2012. Emotion design, emotional design, emotionalize design: A review on their relationships from a new perspective. *Design Journal* 15, 1 (2012), 9–32.
- [60] Wendy Hollway and Tony Jefferson. 2000. *Doing Qualitative Research Differently: Free Association, Narrative and the Interview Method*. Sage.
- [61] Kristina Höök. 2008. Affective loop experiences—What are they? In *Persuasive Technology: 3rd International Conference, Proceedings (PERSUASIVE '08)*. Springer, 1–12.
- [62] Eva Hudlicka. 2008. Affective computing for game design. In *Proceedings of the 4th International North American Conference on Intelligent Games and Simulation*. McGill University, Montreal, Canada, 5–12.
- [63] Eva Hudlicka. 2009. Affective game engines: Motivation and requirements. In *Proceedings of the 4th International Conference on Foundations of Digital Games*. 299–306.
- [64] Eva Hudlicka. 2011. Guidelines for designing computational models of emotions. *International Journal of Synthetic Emotions (IJSE)* 2, 1 (2011), 26–79.
- [65] Robin Hunnicke, Marc LeBlanc, and Robert Zubek. 2004. MDA: A formal approach to game design and game research. In *Proceedings of the AAAI Workshop on Challenges in Game AI*, Vol. 4. 1722.
- [66] Toshie Imada and Phoebe C. Ellsworth. 2011. Proud americans and lucky japanese: Cultural differences in appraisal and corresponding emotion. *Emotion* 11, 2 (2011), 329.
- [67] Katherine Isbister. 2016. *How Games Move Us: Emotion by Design*. MIT Press.
- [68] Carroll E. Izard. 1992. Basic emotions, relations among emotions, and emotion-cognition relations. *Psychol Rev* 99, 3 (1992), 561–565.
- [69] Petar Jerčić, Philipp J. Astor, Marc Adam, Olle Hilborn, Kristina Schaff, Craig Lindley, Charlotte Sennersten, and Jeanette Eriksson. 2012. A serious game using physiological interfaces for emotion regulation training in the context of financial decision-making. In *20th European Conference on Information Systems (ECIS '12)*. AIS Electronic Library (AISeL), 1–14.
- [70] Mijin Kim and Young Yim Doh. 2017. Computational modeling of players' emotional response patterns to the story events of video games. *IEEE Transactions on Affective Computing* 8, 2 (2017), 216–227.
- [71] Sylvia D. Kreibitz. 2010. Autonomic nervous system activity in emotion: A review. *Biological Psychology* 84, 3 (2010), 394–421.
- [72] Annakaisa Kultima. 2015. Game design research. In *Proceedings of the 19th International Academic Mindtrek Conference*. 18–25.
- [73] Peter Kuppens. 2019. Improving theory, measurement, and reality to advance the future of emotion research. *Cognition and Emotion* 33, 1 (2019), 20–23.
- [74] Petri Lankoski. 2012. Computer games and emotions. In *The Philosophy of Computer Games*. Springer, 39–55.
- [75] Effie L-C. Law and Paul Van Schaik. 2010. Modelling user experience—An agenda for research and practice. *Interacting with Computers* 22, 5 (2010), 313–322.
- [76] Richard S. Lazarus. 1968. Emotions and adaptation: Conceptual and empirical relations. In *Nebraska Symposium on Motivation*. University of Nebraska Press.
- [77] Richard S. Lazarus. 1991. *Emotion and Adaptation*. Oxford University Press on Demand.
- [78] Nicole Lazzaro. 2009. Why we play: Affect and the fun of games. *Human-computer Interaction: Designing for Diverse Users and Domains* 155 (2009), 679–700.
- [79] Joseph E. LeDoux. 2000. Emotion circuits in the brain. *Annual Review of Neuroscience* 23, 1 (2000), 155–184.
- [80] Howard Leventhal and Klaus Scherer. 1987. The relationship of emotion to cognition: A functional approach to a semantic controversy. *Cognition and Emotion* 1, 1 (1987), 3–28.

- [81] Ling Li and James Campbell. 2010. Emotion modeling and interaction of NPCs in virtual simulation and games. *International Journal of Virtual Reality* 9, 4 (2010), 1–6.
- [82] Jia Zheng Lim, James Mountstephens, and Jason Teo. 2020. Emotion recognition using eye-tracking: Taxonomy, review and current challenges. *Sensors* 20, 8 (2020), 2384.
- [83] Changchun Liu, Pramila Agrawal, Nilanjan Sarkar, and Shuo Chen. 2009. Dynamic difficulty adjustment in computer games through real-time anxiety-based affective feedback. *International Journal of Human-computer Interaction* 25, 6 (2009), 506–529.
- [84] Jianqiu Mao, Ben Horan, Helen Forbes, Stephen Smilevski, Tracey Bucknall, Cate Nagle, Diane Phillips, and Ian Gibson. 2017. Application of emotional design to the form redesign of a midwifery training aid. In *The International Conference on Design and Technology*. KnowledgeE, 44–50.
- [85] Iris B. Mauss and Michael D. Robinson. 2009. Measures of emotion: A review. *Cognition and Emotion* 23, 2 (2009), 209–237.
- [86] John McCarthy and Peter Wright. 2004. Technology as experience. *Interactions* 11, 5 (2004), 42–43.
- [87] Elisa D. Mekler, Julia Ayumi Bopp, Alexandre N. Tuch, and Klaus Opwis. 2014. A systematic review of quantitative studies on the enjoyment of digital entertainment games. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*. 927–936.
- [88] Wafa Mellouk and Wahida Handouzi. 2020. Facial emotion recognition using deep learning: Review and insights. *Procedia Computer Science* 175 (2020), 689–694.
- [89] Batja Mesquita and Michael Boiger. 2014. Emotions in context: A sociodynamic model of emotions. *Emotion Review* 6, 4 (2014), 298–302.
- [90] Matthew K. Miller and Regan L. Mandryk. 2016. Differentiating in-game frustration from at-game frustration using touch pressure. In *Proceedings of the 2016 ACM International Conference on Interactive Surfaces and Spaces*. 225–234.
- [91] Agnes Moors. 2010. Automatic constructive appraisal as a candidate cause of emotion. *Emotion Review* 2, 2 (2010), 139–156.
- [92] Agnes Moors, Phoebe C. Ellsworth, Klaus R. Scherer, and Nico H. Frijda. 2013. Appraisal theories of emotion: State of the art and future development. *Emotion Review* 5, 2 (2013), 119–124.
- [93] Lennart Erik Nacke, Michael Kalyn, Calvin Lough, and Regan Lee Mandryk. 2011. Biofeedback game design: Using direct and indirect physiological control to enhance game interaction. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*. 103–112.
- [94] Yiing Ng, Chee Weng Khong, and Robert Jeyakumar Nathan. 2018. Evaluating affective user-centered design of video games using qualitative methods. *International Journal of Computer Games Technology* 2018 (2018), 1–13.
- [95] Pedro Nogueira, Rui Rodrigues, Eugénio Oliveira, and Lennart Nacke. 2013. Guided emotional state regulation: Understanding and shaping players’ affective experiences in digital games. In *Proceedings of the AAAI Conference on Artificial Intelligence and Interactive Digital Entertainment*, Vol. 9.
- [96] Pedro A. Nogueira, Vasco Torres, Rui Rodrigues, Eugénio Oliveira, and Lennart E. Nacke. 2016. Vanishing scares: Biofeedback modulation of affective player experiences in a procedural horror game. *Journal on Multimodal User Interfaces* 10, 1 (2016), 31–62.
- [97] Donald A. Norman. 2004. *Emotional Design: Why We Love (or Hate) Everyday Things*. Basic Civitas Books.
- [98] Jaak Panksepp. 2004. *Affective Neuroscience: The Foundations of Human and Animal Emotions*. Oxford University Press.
- [99] Supavich Fone Pengnate and Rathindra Sarathy. 2017. An experimental investigation of the influence of website emotional design features on trust in unfamiliar online vendors. *Computers in Human Behavior* 67 (2017), 49–60.
- [100] Bernard Perron. 2005. A cognitive psychological approach to gameplay emotions. In *DiGRA Conference* (2005).
- [101] Bernard Perron. 2012. *Silent Hill: The Terror Engine*. University of Michigan Press.
- [102] Rosalind W. Picard. 2000. *Affective Computing*. MIT Press.
- [103] Thammathip Piumsomboon, Gun Lee, Robert W. Lindeman, and Mark Billingham. 2017. Exploring natural eye-gaze-based interaction for immersive virtual reality. In *2017 IEEE Symposium on 3D User Interfaces (3DUI '17)*. IEEE, 36–39.
- [104] Jonathan Posner, James A. Russell, and Bradley S. Peterson. 2005. The circumplex model of affect: An integrative approach to affective neuroscience, cognitive development, and psychopathology. *Development and Psychopathology* 17, 3 (2005), 715.
- [105] CD Project. 2020. *Cyberpunk 77*. [Game].
- [106] Rajagopal Raghunathan, Michel T. Pham, and Kim P. Corfman. 2006. Informational properties of anxiety and sadness, and displaced coping. *Journal of Consumer Research* 32, 4 (2006), 596–601.
- [107] Rainer Reisenzein, Eva Hudlicka, Mehdi Dastani, Jonathan Gratch, Koen Hindriks, Emiliano Lorini, and John-Jules Ch Meyer. 2013. Computational modeling of emotion: Toward improving the inter-and intradisciplinary exchange. *IEEE Transactions on Affective Computing* 4, 3 (2013), 246–266.

- [108] Raquel Robinson, Katelyn Wiley, Amir Rezaeivahdati, Madison Klarkowski, and Regan L. Mandryk. 2020. “Let’s get physiological, physiological!” A systematic review of affective gaming. In *Proceedings of the Annual Symposium on Computer-human Interaction in Play*. 132–147.
- [109] Karen Robson, Kirk Plangger, Jan H. Kietzmann, Ian McCarthy, and Leyland Pitt. 2015. Is it all a game? Understanding the principles of gamification. *Business Horizons* 58, 4 (2015), 411–420.
- [110] Anna Marie Ruef and Robert W. Levenson. 2007. Continuous measurement of emotion: The affect rating dial. In *Handbook of Emotion Elicitation and Assessment*, James A. Coan and John J. B. Allen (Eds.). Oxford University Press, New York, NY, 286–297.
- [111] James A. Russell. 1980. A circumplex model of affect. *Journal of Personality and Social Psychology* 39, 6 (1980), 1161.
- [112] James A. Russell. 1983. Pancultural aspects of the human conceptual organization of emotions. *Journal of Personality and Social Psychology* 45, 6 (1983), 1281.
- [113] Richard M. Ryan, C. Scott Rigby, and Andrew Przybylski. 2006. The motivational pull of video games: A self-determination theory approach. *Motivation and Emotion* 30, 4 (2006), 344–360.
- [114] Katie Salen, Katie Salen Tekinbaş, and Eric Zimmerman. 2004. *Rules of Play: Game Design Fundamentals*. MIT Press.
- [115] David Sander, Didier Grandjean, and Klaus R. Scherer. 2005. A systems approach to appraisal mechanisms in emotion. *Neural Networks* 18, 4 (2005), 317–352.
- [116] Nikolaos Savva, Alfonsina Scarinzi, and Nadia Bianchi-Berthouze. 2012. Continuous recognition of player’s affective body expression as dynamic quality of aesthetic experience. *IEEE Transactions on Computational Intelligence and AI in Games* 4, 3 (2012), 199–212.
- [117] Jesse Schell. 2008. *The Art of Game Design: A Book of Lenses*. CRC Press.
- [118] Klaus R. Scherer. 1993. Neuroscience projections to current debates in emotion psychology. *Cognition & Emotion* 7, 1 (1993), 1–41.
- [119] Klaus R. Scherer. 2019. Studying appraisal-driven emotion processes: Taking stock and moving to the future. *Cognition and Emotion* 33, 1 (2019), 31–40.
- [120] Klaus R. Scherer et al. 2000. Psychological models of emotion. *Neuropsychology of Emotion* 137, 3 (2000), 137–162.
- [121] Emery Schubert. 1999. Measuring emotion continuously: Validity and reliability of the two-dimensional emotion-space. *Australian Journal of Psychology* 51, 3 (1999), 154–165.
- [122] Yoonas A. Sekhavat, Milad Jafari Sisi, and Samad Roohi. 2021. Affective interaction: Using emotions as a user interface in games. *Multimedia Tools and Applications* 80, 4 (2021), 5225–5253.
- [123] Noor Shaker, Stylianos Asteriadi, Georgios N. Yannakakis, and Kostas Karpouzis. 2013. Fusing visual and behavioral cues for modeling user experience in games. *IEEE Transactions on Cybernetics* 43, 6 (2013), 1519–1531.
- [124] Noor Shaker, Georgios Yannakakis, and Julian Togelius. 2010. Towards automatic personalized content generation for platform games. In *Proceedings of the AAAI Conference on Artificial Intelligence and Interactive Digital Entertainment*, Vol. 5.
- [125] Richard M. Shiffrin and Walter Schneider. 1977. Controlled and automatic human information processing: II. Perceptual learning, automatic attending and a general theory. *Psychological Review* 84, 2 (1977), 127.
- [126] Erika H. Siegel, Molly K. Sands, Wim Van den Noortgate, Paul Condon, Yale Chang, Jennifer Dy, Karen S. Quigley, and Lisa Feldman Barrett. 2018. Emotion fingerprints or emotion populations? A meta-analytic investigation of autonomic features of emotion categories. *Psychological Bulletin* 144, 4 (2018), 343.
- [127] Aaron Smuts. 2005. Are video games art? *Contemporary Aesthetics (Journal)* 3, 1 (2005), 6.
- [128] Jose L. Soler-Dominguez, Jorge D. Camba, Manuel Contero, and Mariano Alcañiz. 2017. A proposal for the selection of eye-tracking metrics for the implementation of adaptive gameplay in virtual reality based games. In *International Conference on Virtual, Augmented and Mixed Reality*. Springer, 369–380.
- [129] Mohammad Soleymani, Maja Pantic, and Thierry Pun. 2011. Multimodal emotion recognition in response to videos. *IEEE Transactions on Affective Computing* 3, 2 (2011), 211–223.
- [130] Andreas Sonderegger and Juergen Sauer. 2010. The influence of design aesthetics in usability testing: Effects on user performance and perceived usability. *Applied Ergonomics* 41, 3 (2010), 403–410.
- [131] Petra Sundström. 2005. *Exploring the Affective Loop*. Ph.D. Dissertation.
- [132] Jonathan Sykes and Simon Brown. 2003. Affective gaming: Measuring emotion through the gamepad. In *CHI’03 Extended Abstracts on Human Factors in Computing Systems*. 732–733.
- [133] David Thue, Vadim Bulitko, Marcia Spetch, and Trevon Romanuik. 2010. Player agency and the relevance of decisions. In *Joint International Conference on Interactive Digital Storytelling*. Springer, 210–215.
- [134] Tim Tijs, Dirk Brokken, and Wijnand IJsselstein. 2008. Creating an emotionally adaptive game. In *International Conference on Entertainment Computing*. Springer, 122–133.
- [135] Silvan S. Tomkins. 1984. Affect theory. In *Emotion in the Human Face*, P. Ekman (Ed.). (2nd ed.), Cambridge University Press, New York, NY, 353–395.

- [136] Gustavo Fortes Tondello. 2019. *Dynamic Personalization of Gameful Interactive Systems*. University of Waterloo, Waterloo, Canada PhD Thesis.
- [137] April Tyack and Elisa D. Mekler. 2021. Off-peak: An examination of ordinary player experience. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*, 1–12.
- [138] Mark A. Urista, Qingwen Dong, and Kenneth Day. 2009. Explaining why young adults use MySpace and Facebook through uses and gratifications theory. *Human Communication* 12, 2 (2009), 215–229.
- [139] Vanus Vachiratamporn, Roberto Legaspi, Koichi Moriyama, Ken-ichi Fukui, and Masayuki Numao. 2015. An analysis of player affect transitions in survival horror games. *Journal on Multimodal User Interfaces* 9, 1 (2015), 43–54.
- [140] David Watson, Lee Anna Clark, and Auke Tellegen. 1988. Development and validation of brief measures of positive and negative affect: The PANAS scales. *Journal of Personality and Social Psychology* 54, 6 (1988), 1063.
- [141] Lisa E. Williams, Jonathan A. Oler, Andrew S. Fox, Daniel R. McFarlin, Gregory M. Rogers, Maria A. L. Jesson, Richard J. Davidson, Daniel S. Pine, and Ned H. Kalin. 2015. Fear of the unknown: Uncertain anticipation reveals amygdala alterations in childhood anxiety disorders. *Neuropsychopharmacology* 40, 6 (2015), 1428–1435.
- [142] Christine D. Wilson-Mendenhall, Lisa Feldman Barrett, and Lawrence W. Barsalou. 2013. Neural evidence that human emotions share core affective properties. *Psychological Science* 24, 6 (2013), 947–956.
- [143] Peter Wright and John McCarthy. 2008. Empathy and experience in HCI. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*. 637–646.
- [144] Wilhelm Max Wundt. 1904. *Principles of Physiological Psychology*. Vol. 1. Sonnenschein.
- [145] Georgios N. Yannakakis, Héctor P. Martínez, and Arnav Jhala. 2010. Towards affective camera control in games. *User Modeling and User-Adapted Interaction* 20, 4 (2010), 313–340.
- [146] Georgios N. Yannakakis and Ana Paiva. 2014. Emotion in games. *Handbook on Affective Computing* 2014 (2014), 459–471.
- [147] Georgios N. Yannakakis and Julian Togelius. 2011. Experience-driven procedural content generation. *IEEE Transactions on Affective Computing* 2, 3 (2011), 147–161.
- [148] Veronica Zammitto. 2018. Games user research as part of the development process in the game industry: Challenges and best practices. In *Games User Research*, Anders Drachen, Pejman Mirza-Babaei, and Lennart E. Nacke (Eds.). Oxford University Press, Oxford, UK. 15–30.
- [149] Dolf Zillmann. 2003. Theory of affective dynamics: Emotions and moods. In *Communication and Emotion*. Routledge, 533–567.

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